

1 UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

2 DOCTORAL THESIS

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4 **The measurement of Higgs**
5 **production cross section in the**
6 **four-lepton final state at CMS**

7

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9 *A thesis submitted in fulfillment of the requirements*
10 *for the degree of Doctor of Science*

11 *in the*

12 **Institute of High Energy Physics**
13 **Chinese Academy of Sciences**



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Chapter 1

Physics Objects

As described in section ??, that events organized in streams of HLT decisions are eventually transferred to the storage system (Tier-0) of CMS for its permanent storage. That data is in RAW format and needs to be fully reconstructed from this storage system. With potential reconstruction procedure, this RAW data can be associated to physical particles such as simple or general tracks, muons, photons, electrons and jets etc. These reconstructed particles then help to explore *e.g.* Higgs decay to four lepton channel for its properties measurement like spin, parity, cross sections, mass and etc. The reconstruction of the SM Higgs boson in the decay chain $H \rightarrow ZZ \rightarrow 4\ell$ requires very efficient lepton reconstruction and identification in order to be sensitive to a low mass Higgs. In this kinematics region, there is need to attain an optimal reconstruction efficiency keeping misidentified lepton rate as low as possible. Further, the precise measurements on Higgs are lepton kinematics dependent and thus needs high precision momentum measurement.

1.1 Reconstruction of Track and Vertex

Reconstructing a track is procedure of exploiting hits in the detector parts used to estimate the momentum and/or position of a particle producing the hits. It is a challenging task and in the CMS experiment there a dedicated software known as Combinatorial Track Finder (CTF). It uses combinatorial Kalman filter approach for the recognition of patterns and fitting the track.

CTF algorithm repeats in multiple steps to search the track on the basis of its relatively larger p_T and closeness to the interaction point [Chatrchyan:2014fea].

This step is called iterative tracking in which each iteration is done in following steps:

- 158 • **Seeding:** It is to provide with the help of 2 or 3 hits initial track candidates
159 (seeds).
- 160 • **Tracking Finding:** It is to search for the additional hits based on the seed
161 trajectories based on the Kalman filter approach.
- 162 • **Estimate of trajectory:** It is to use the Kalman-filter and smoother to estimate
163 the best parameters of each trajectory.
- 164 • **Track Selection:** Based on the quality of the trajectories, those tracks which
165 do not originate from some charged particle are discarded.

166 It is repeated for 6 times which are done to figure out the tracks originating
167 from outside of the luminous region of the proton proton collisions.

168
169 The completely reconstructed tracks give the information of the proton-proton
170 interaction vertices. Reconstruction of the primary-vertex is done in three steps:

- 171 • **Selection of Tracks:** It is done by setting requirements on significance of im-
172 pact parameter (SIP) from beam spot, number of hits in ITS and normalized
173 χ^2 .
- 174 • **Track Clustering:** It is done using information of transverse component of
175 the impact parameter with respect to the beam spot. The clustering algo-
176 rithm is capable to reconstruct any number of proton proton interaction in
177 the bunch crossing. This performed using deterministic annealing algorithm
178 (DA). This algorithm clusters the tracks into groups which can be associated
179 to a single vertex.
- 180 • **Fitting of the track:** It is performed over the tracks in each cluster using
181 "adaptive vertex fitter" to get the position of the primary vertex (the best
182 vertex parameters). It needs a procedure to link a primary vertex with the
183 hard interaction. Since in hard interaction particles are produced at high p_T ,
184 for example in $Z \rightarrow \mu^+ \mu^-$ process, the vertices can be associated with sum of
185 p_T^2 of the reconstructed tracks. The vertex with highest Σp_T^2 can be associated
186 to the hard intraction from the multiplicity of primary vertices. Rest of the
187 interactions, other than hard interaction, are called pileup.

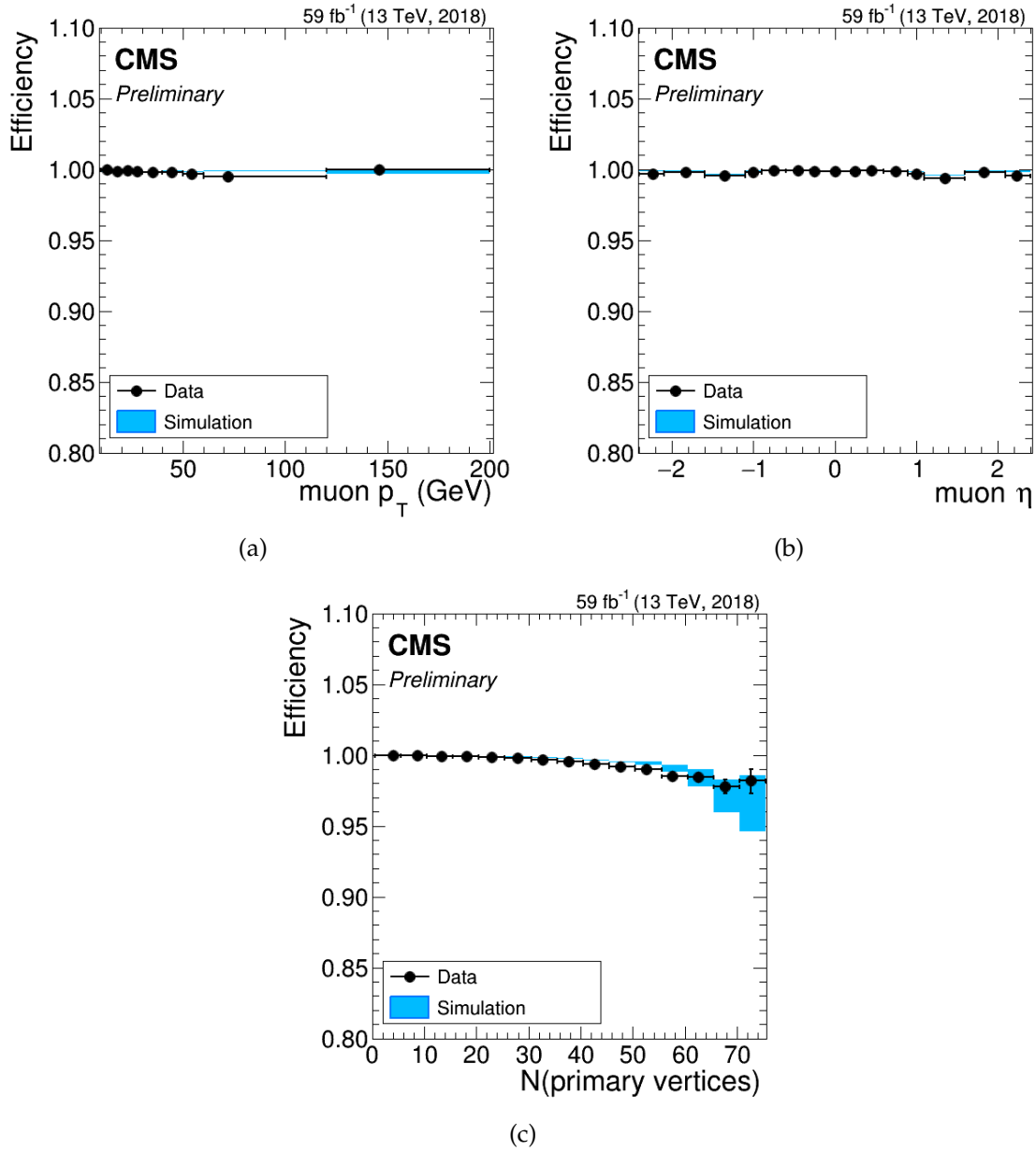


FIGURE 1.1: The tracking efficiency as a function of p_T pseudo-rapidity and number of primary vertices for the standalone muon for the “all tracks” collection are shown in the sub-figure top left, top right and bottom respectively. Data (black dots) and Madgraph Drell Yan i.e. $Z \rightarrow \mu^+\mu^-$ (light blue rectangles). The uncertainties shown are statistical.

188 The efficiency to reconstruct a lepton track is called the tracking efficiency. For
 189 instance, to reconstruct a muon track in the inner detector is measured using as
 190 probes tracks reconstructed in the muon system alone. Measured muon tracking

efficiency is shown in the Figure. 1.1. It is evident from Figures. 1.1(a) and 1.1(b) that tracking efficiency is nearly 100% in terms of p_T and η ; however, with increase in number of primary vertices (pileup) it decreases a bit as shown in the Figure. 1.1(c).

These efficiencies are measured using “Tag and Probe ” method for which details are described in next Section 1.1.1. This procedure is also applied when measuring the selection efficiencies of electrons and muons.

1.1.1 Tag and Probe Analysis

Simulation studies are important to validate the measurements on data. However, there are possibilities that the studies based on the simulations can be biased due to some insufficient event description and the lack of understanding of the detector. Further, for an analysis we always optimize the the event selection and in an ideal selection, we efficiently eliminate the background events with optimal signal events. Every real selection affects signal efficiencies which should be estimated. These efficiencies can be extracted directly from the simulation, but there is a danger such as bias which can appear as mentioned above. Such bias from MC, can be estimated by measuring the selection efficiencies directly from the experimental data. In high purity environment, one can measure the efficiencies by cut and cout method; however, if significant background is involved one needs to measure the efficiencies by “fit method” i.e. by using the Tag-and-Probe (TnP) method.

The TnP Tool is a generic tool developed to measure any user defined object efficiency (e.g. in this dissertation the reconstruction and/or identification, impact parameter cuts isolation and trigger) from data or MC by exploiting the di-object resonances like Z , J/ψ or Υ [tnp]. Z boson decaying to a lepton pair is considered to be standard candle for the measurement of these efficiencies (except for reconstruction and identification of muons at low p_T (i.e. $p_T < 20$ GeV) where J/ψ resonances are used). Brief description of the procedure is given as under:

- Resonances are reconstructed as pairs with one leg passing a tight selection (called “tag”) and one passing a loose selection (called “probe”).
- According to whatever efficiency to be measured, passing probes (from all probes) are defined if they also pass the selection. Those which do not pass the selection are call “failing probes”.

- Di-leptons mass distributions are built using “tag + passing probes” and “tag + failing probes”, and are separately fit using “signal + background” model.
- The ratio of the signal yields in these two lineshapes give the subject “efficiency”.
- The procedure is repeated in bins of probe variables (i.e. p_T, η and so on) to visualize the efficiencies in response of varying these variables.

An example of a TnP fit is shown in the Figure. 1.2. And schematic workflow of the TnP method while performing the procedure in measurements using the CMS central framework is shown in the Figure. 1.3.

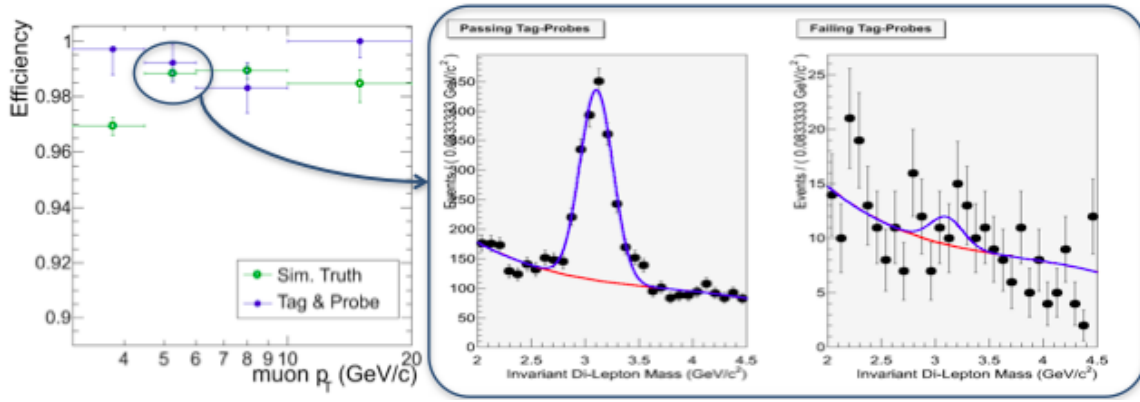


FIGURE 1.2: Efficiency plots in bins of p_T alongwith an example of TnP fit of di-lepton distributions. In the fit plot, black points are data points with statistical uncertainty bars, red lineshape is background fit and blue lineshape is “signal+background” fit in case of “tag + passing probes” (left) and “tag + failing probes” (right)

1.2 Electrons and Photons

1.2.1 Electron and Photon Reconstruction

Both of electron and photon are important ingredients of the analysis upon which the dissertation is based. Both interact electromagnetically and can be reconstructed and identified at CMS very efficiently which is also the requirement physics analyses. Photon being chargeless deposits its energy only in the electromagnetic calorimeter (ECAL), as described in Section. ??, leaving no track in the

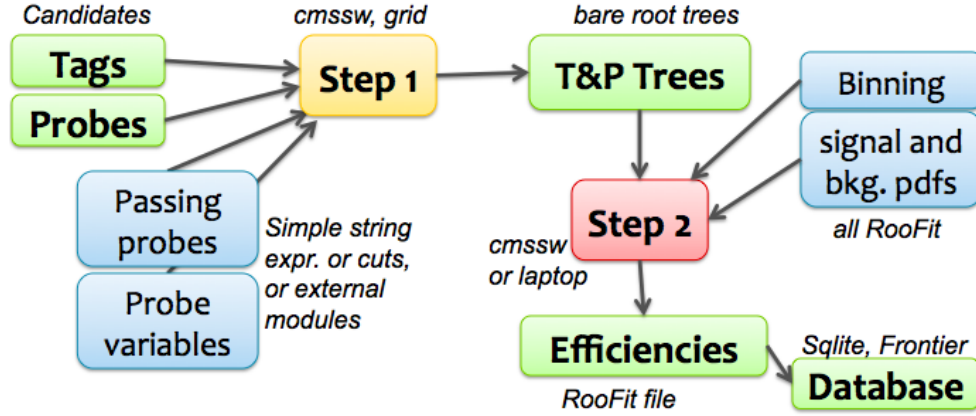


FIGURE 1.3: Tag and Probe workflow example using the central CMS framework.

ITS of CMS. On the other hand, electrons leave tracking information in the ITS and well as in the ECAL. A schematic diagram of the behavior of photon and electron as they fly out of the interaction point from ITS to ECAL is shown in the Figure 1.4.

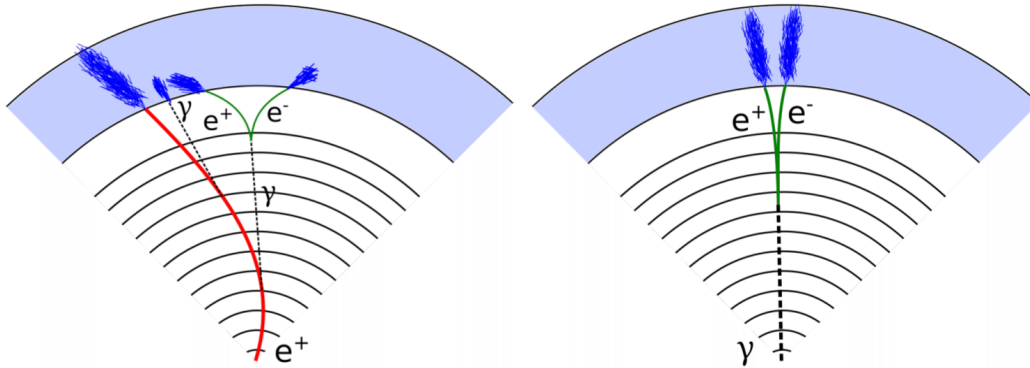


FIGURE 1.4: Pattern of positron (photon) to the detector shown left (right).

There are many ways to identify the true or uncovered photons e.g. by using:

- Tracker, ECAL and HCAL isolation
- Hadronic to electromagnetic ratio
- R_9 ratio

Electrons can be reconstructed and identified when a supercluster is associated to a track in the silicon detector. For track reconstruction of electron, as described in Section 1.1.1, ECAL based seeding is efficient for $p_T > 10$; however, for lower p_T Tracker based seeding (that makes use of boosted decision tree (BDT) for preselection of tracker clusters) is more efficient. p_T of the electron is measured from these tracker tracks produced when it pass through it. Being charged particle, it has curved track under the influence of superconducting magnet due to which it radiates to photons. Therefore, Gaussian Sum Filter (GSF) algorithm, which takes into account the bremsstrahlung losses, is used to reconstruct the track.

Preselection of the electron candidates is made through loose cuts on track-cluster matching observables. This done to maintain highest possible selection efficiency of electrons and to reject QCD background significantly. For the analysis, electrons are subjected to have $p_T^e > 7$ GeV within $|\eta^e| < 2.5$, and satisfy a loose primary vertex constraints on longitudinal and transvers components of impact parameter *viz.* $d_{xy} < 0.5$ cm and $d_z < 1$ cm respectively. Such electrons are called **loose electrons**. Details of electron reconstruction can be explored in Ref. [1].

The data-MC discrepancy is corrected using scale factors as is done for the electron selection with data efficiencies measured through technique outlined later (see Section 1.2.5). These studies for reconstructions are carried out by the EGM POG and the results are only summarised here.

The electron reconstruction scale factors are shown Fig. 1.5 and are applied as a function of electron p_T and super cluster η .

1.2.2 Electron Identification and Isolation

Electron identification and isolation is performed new Gradient Boosted Decision Tree (GBDT), a multivariate classifier algorithm. It exploits the informations from electromagnetic cluster, cluster-track matching, the variables which exclusively depend on tracking measurements and as well as particle flow isolation sums. The list of observables used can be seen in the Table 1.1.

For both signal and the background, the classifier is trained on 2017 MC samples of Drell-Yan(DY) plus jets. Fig. 1.6 shows the output of the multiclassifier discriminant for prompt electrons from DY events and fakes from jets in DY events.

The impact of the transition from the TMVA(v1) to the xgboost(v2) training library is shown in Fig. 1.7. The working points shown are adopted to get the

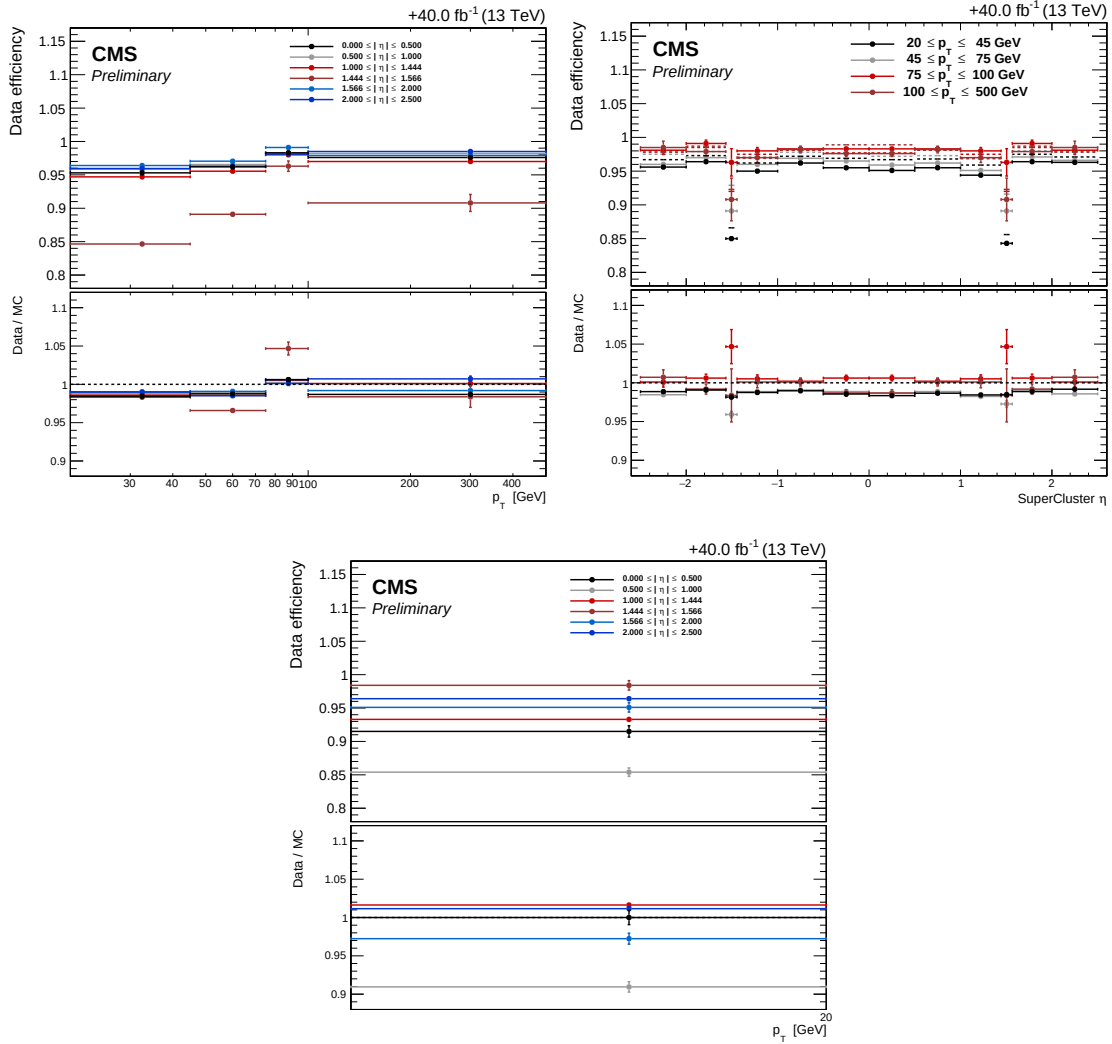


FIGURE 1.5: Reconstruction efficiencies of electron in data versus p_T (left), η (right) for electrons having $p_T > 20$ GeV (top) and $p_T < 20$ GeV (bottom) with corresponding data/MC scale factors as provided by the EGM POG.

same signal efficiency as the cut on MVA ID and a cut on the PF isolation, in each p_T and η bin.

Table 1.2 lists the cut values applied to the BDT score for the chosen working point. For the analysis, loose electrons have to pass this MVA identification and isolation working point.

observable type	observable name
6*cluster shape	RMS value of spectrum of energy-crystal number along η and ϕ ; $\sigma_{i\eta i\eta}$, $\sigma_{i\phi i\phi}$ width of super cluster along η and ϕ ratio of hadronic energy behind the supercluster of electron to supercluster energy, H/E circularity $(E_{5\times 5} - E_{5\times 1})/E_{5\times 5}$ sum of the seed and adjacent crystal over the super cluster energy R_9 for the endcap training bins: pre-shower energy fraction E_{PS}/E_{raw}
2*track-cluster matching	energy-momentum agreement E_{tot}/p_{in} , E_{ele}/p_{out} , $1/E_{tot} - 1/p_{in}$ position matching $\Delta\eta_{in}$, $\Delta\phi_{in}$, $\Delta\eta_{seed}$
5*tracking	fractional loss of momentum loss $f_{brem} = 1 - p_{out}/p_{in}$ number of hits of KF and the GSF track N_{KF} , N_{GSF} ($\cdot$) reduced χ^2 of KF and the GSF track χ^2_{KF} , χ^2_{GSF} number of expected innre hits (missing) ($\cdot$) probability transformation in conversion vertex fit χ^2 ($\cdot$)
3*isolation	Particle Flow photon isolation sum ($\cdot$) Particle Flow charged hadrons isolation sum ($\cdot$) Particle Flow neutral hadrons isolation sum ($\cdot$)
1*For PU-resilience	Mean energy density in the event: ρ ($\cdot$)

TABLE 1.1: Overview of variables used at inputs to identification classifier. The Variables not used in the Run I MVA are specified with ($\cdot$).

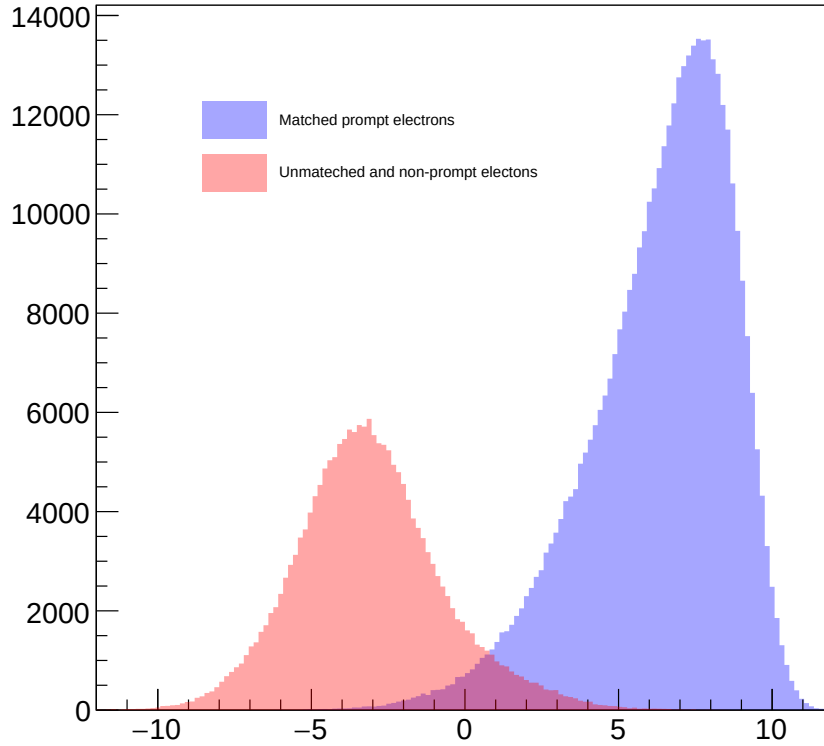


FIGURE 1.6: Output of the multiclassifier discriminant for prompt electrons matched to truth electrons from Z decay (blue) and for fakes (red). Events are all taken from DY events MC sample.

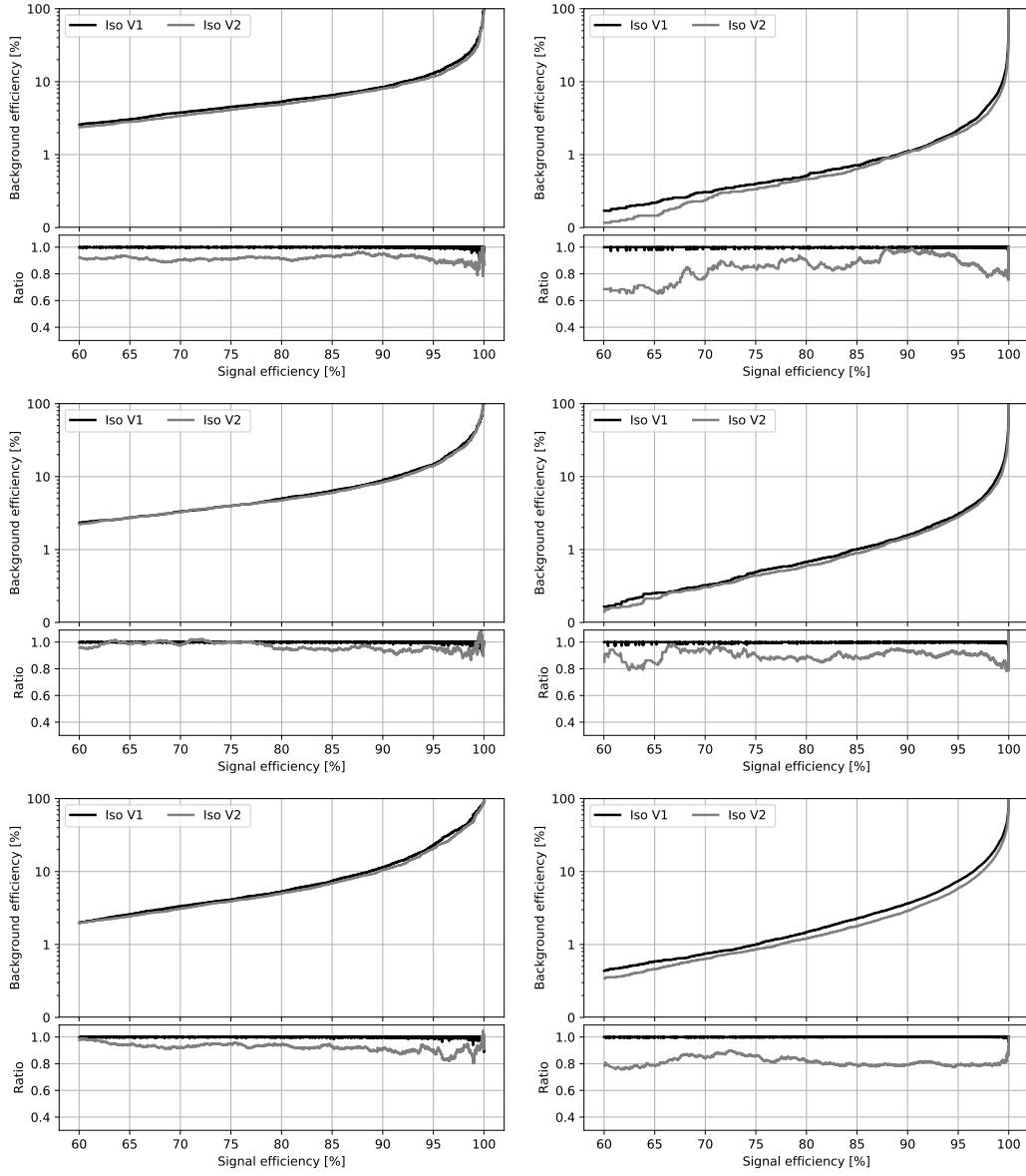


FIGURE 1.7: The receiver operating characteristic curves (background efficiency vs signal efficiency) of the MVA's trained on the 2017 simulation and applied on the 2018 simulation. The training combines identification and isolation variables. Performance are shown for electrons with $5 < p_T < 10 \text{ GeV}$ (left), $p_T > 10 \text{ GeV}$ (right) and $|\eta| < 0.8$ (top), $0.8 < |\eta| < 1.479$ (middle) and $|\eta| > 1.479$ (bottom). V1 and V2 versions of training are compared, exploiting TMVA and xgboost training libraries respectively.

		$ \eta < 0.8$	
		Cut on BDT score	Signal eff. Background eff.
$5 < p_T < 10 \text{ GeV}$	1.264	81.04%	4.4%
$p_T > 10 \text{ GeV}$	2.364	97.1%	2.9%
		$0.8 < \eta < 1.479$	
		Cut on BDT score	Signal eff. Background eff.
$5 < p_T < 10 \text{ GeV}$	1.178	79.3%	4.6%
$p_T > 10 \text{ GeV}$	2.078	96.3%	3.6%
		$ \eta > 1.479$	
		Cut on BDT score	Signal eff. Background eff.
$5 < p_T < 10 \text{ GeV}$	1.330	72.97%	3.6%
$p_T > 10 \text{ GeV}$	1.080	95.7%	6.7%

TABLE 1.2: Minimum value of BDT score for electron to pass identification, together with the corresponding signal and the background efficiencies.

1.2.3 Electron Impact Parameter Selection

For lepton to be consistent with some common primary vertex, we require that they have an associated track that corresponds a small impact parameter from the event primary vertex. We use the significance of impact parameter to the vertex of an event, i.e. $|\text{SIP}_{3\text{D}}| = \frac{\text{IP}}{\sigma_{\text{IP}}}$, where IP being three dimensional lepton impact parameter w.r.t the primary interaction vertex, and the σ_{IP} being the uncertainty associated to this. Hence, a lepton satisfying $|\text{SIP}_{3\text{D}}| < 4$ is regarded as "primary lepton".

1.2.4 Electron Energy Calibrations

To measure the electron energy scale, a Crystal-ball function is used to fit dielectron mass distribution around the nominal Z boson peak.

It is measured in bins of p_T and $|\eta|$ for corrections to be applied on $Z \rightarrow ee$ sample to align the dielectron mass spectrum in data to that in MC, and to minimize its width.

The resolution of $Z \rightarrow ee$ mass in the Monte Carlo is made to match with data by applying a pseudorandom Gaussian smearing to electron energies, with varying parameters of Gaussian in p_T and $|\eta|$ bins. This has effect of convoluting the electron energy spectrum with a Gaussian.

The energy scale for the full 2018 dataset is shown in Fig. 1.8(a) and agrees decently with the MC with the corrections released by EGAMMA POG.

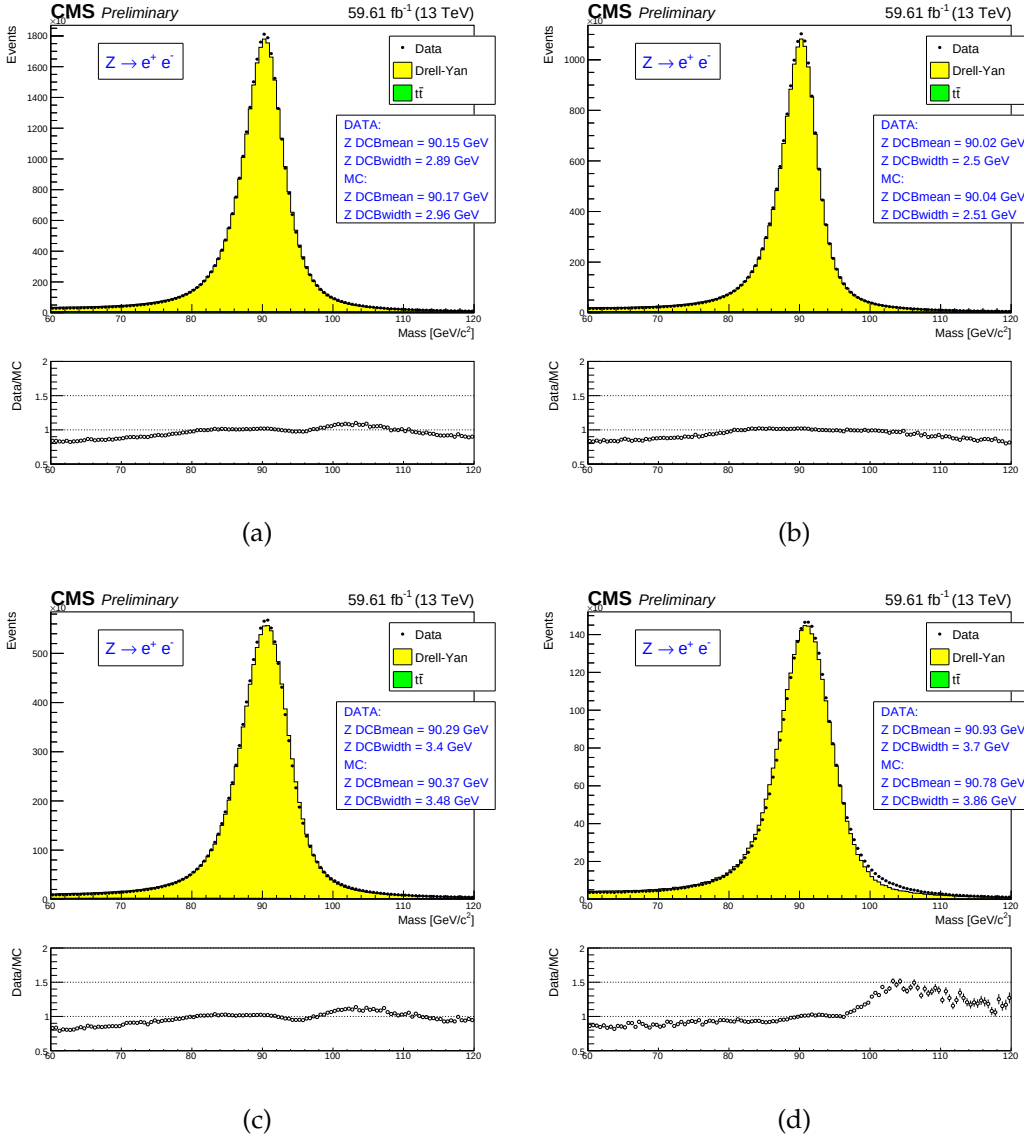


FIGURE 1.8: (a) electron energy scale measured in the $Z \rightarrow ee$ control region for all electrons, (b) for both electrons in barrel, (c) for one electron in barrel and other in endcaps, and for both of the electrons in endcaps (d). In these figures, the results based on fits using Crystall-ball are shown.

1.2.5 Electron Efficiency Measurements

The Tag-and-Probe study was performed on the EGM dataset using the golden JSON of 58.83, 41.4 and 35.9 fb⁻¹ respectively for 2018, 2017 and 2016. More details on this technique can be found in Ref. [2].

Tag electrons need to satisfy the following quality requirements:

- trigger matched to HLT_Ele32_WPTight_Gsf_L1DoubleEG_v*
- $p_T > 30$ GeV, super cluster (SC) $\eta < 2.17$

For the bin between 7 and 20 GeV, additional criteria are required: the tag has to pass a cut on the MVA ID > 0.92 , $\sqrt{(2 * PFMET * p_T(tag) * (1 - \cos(\phi_{MET} - \phi_{tag})))} < 45 \text{ GeV}$.

Probe electrons only need to be reconstructed as GsfElectron while the FSR recovery algorithm is not applied in efficiency measurement.

The nominal MC efficiencies are evaluated from the LO MadGraph Drell-Yan, while the NLO systematics use the MadGraph_AMCatNLO sample.

For the efficiency measurements a template fit is used. The m_{ee} signal shape of the passing and failing probes is taken from MC and convoluted with a Gaussian. The data is then fitted with the convoluted MC template and a CMSShape (an Error-function with a one-sided exponential tail). This change follows from the usage of the T&P tool developed by the EGM POG.

The measurement of electron selection efficiency is made as a function of the probe electron p_T and its SC η . For electrons falling in the ECAL gaps, it is measured separately. Figure 1.9 and ?? shows the p_T and η turn-on curves measured in data.

The EGM recommendations about evaluation of Tag-and-Probe uncertainties on efficiency measurements are followed. Specifically, we consider

- Variation of the signal shape from a MC shape to an analytic shape (Crystal Ball) fitted to the MC
- Use of analytic function (Crystal Ball) to fit the signal as alternative of MC template
- Using an NLO MC sample for signal templates

Assessment of total uncertainty for measurement of the scale factors is taken as quadratic sum of statistical uncertainties (returned from the fit) and the aforementioned systematic uncertainties. Overall electron SFs and the associated total uncertainties can be seen in the Figure. 1.11.

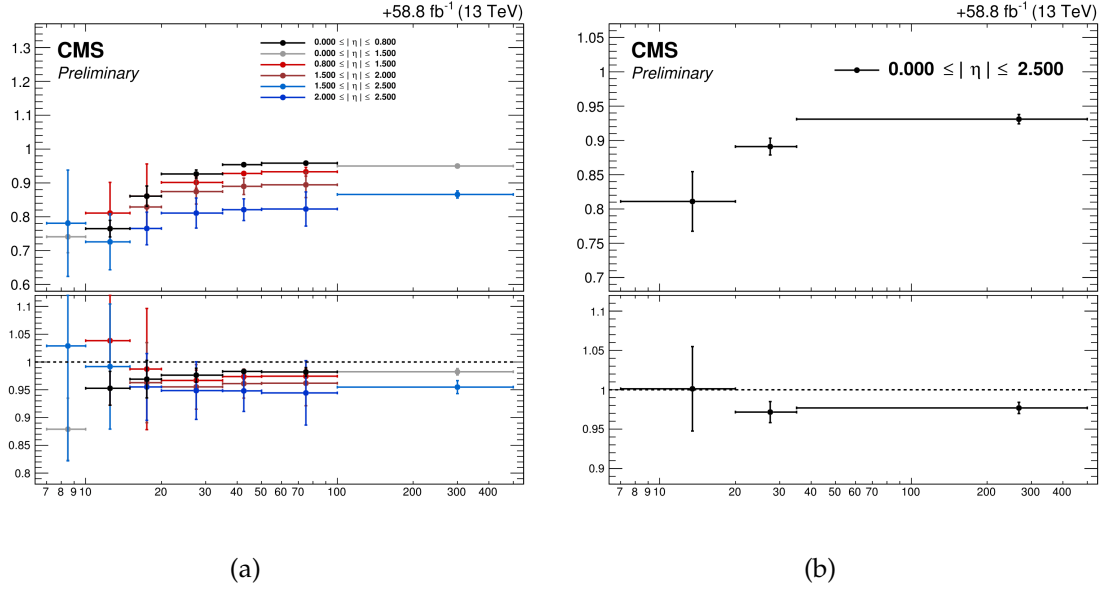


FIGURE 1.9: Electron selection efficiencies vs p_T measured using the Tag-and-Probe technique, non-gap electrons (left) and gap electrons (right), together with the corresponding data/MC ratio bottom part of each plot.

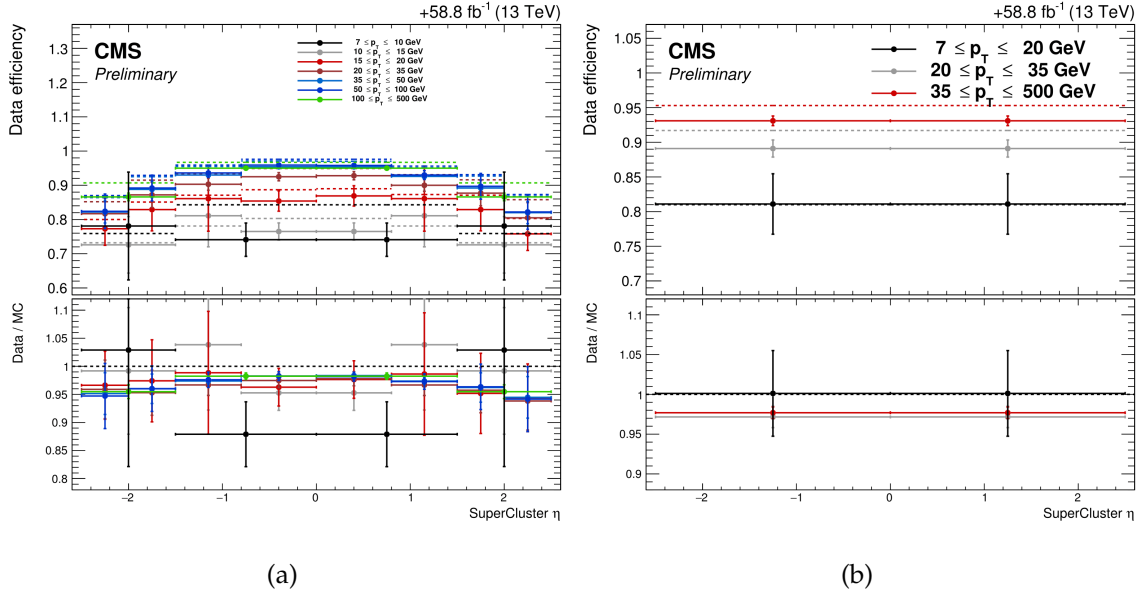


FIGURE 1.10: Electron selection efficiencies vs η , non-gap electrons (left) and gap electrons (right), together with the corresponding data/MC ratio at the bottom part of each plot.

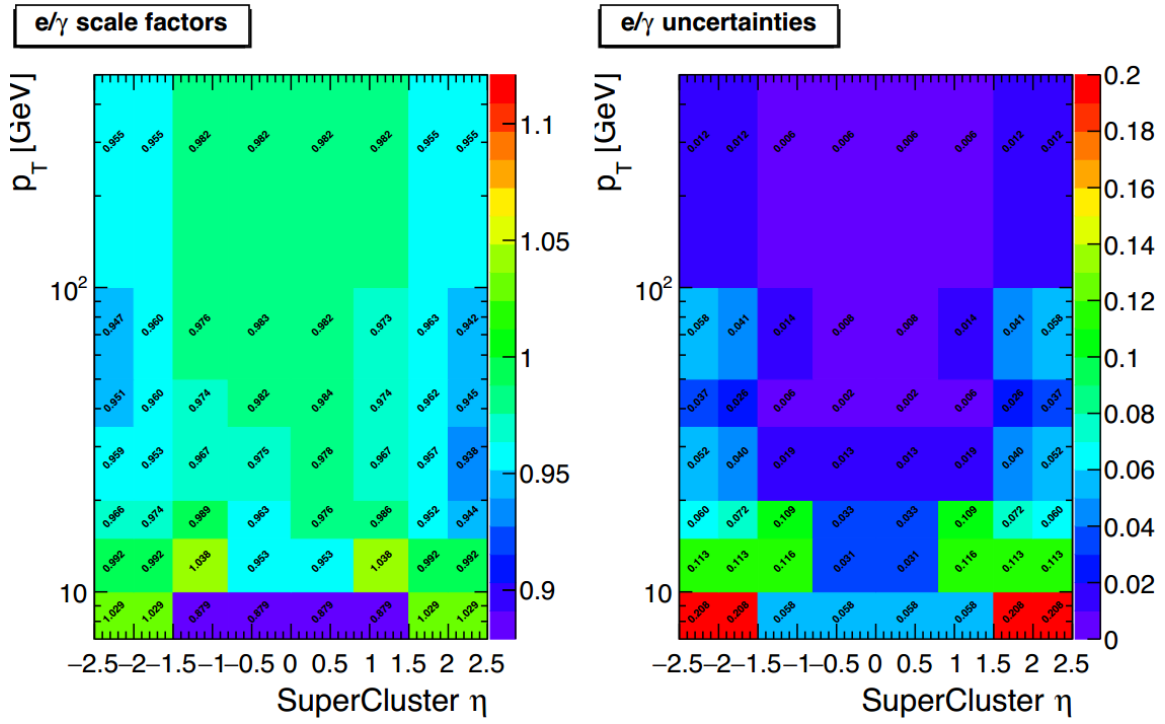


FIGURE 1.11: Overall electron SFs (left) and associated uncertainties (right) in two dimensional display as function of p_T and supercluster η .

1.3 Muons

1.3.1 Muon Reconstruction and Identification

Muons can be reconstructed as global muon which exploits the information from muon track in muon system and a tracker track. It can also be reconstructed by matching the tracker track with its track stubs (i.e. with at least one the muon track segment created by CSC or DT hits).

As per requirement of analysis, we define **loose muons** as the muons that satisfy $p_T > 5$, $|\eta| < 2.4$, $dxy < 0.5$ cm, $dz < 1$ cm, where dxy and dz are defined w.r.t. the PV and using the 'muonBestTrack'. Muons have to be reconstructed by either the Global Muon or Tracker Muon algorithm. Standalone Tracks of muon reconstructed only in the muon system are rejected. Muons with `muonBestTrackType==2` (standalone) are discarded even if they are marked as global or tracker muons. More details on the reconstruction of muons can be seen in Ref. [2].

Loose muons with p_T below 200 GeV are considered identified muons if they also pass the PF muon ID (note that the naming convention used for these IDs differs from the muon POG naming scheme, in which the “tight ID” used here is called the “loose ID”). Loose muons with p_T above 200 GeV are considered identified muons if they pass the PF ID or the Tracker High- p_T ID, the definition of which is shown in Table 1.3. This relaxed definition is used to increase signal efficiency for the high-mass search. When a very heavy resonance decays to two bosons, both bosons will be very boosted. In the lab frame, the leptons originating from decay of a highly boosted Z will be nearly collinear, and the PF ID loses efficiency for muons separated by approximately $\Delta R < 0.4$, which roughly corresponds to muons originating from bosons with $p_T > 500$ GeV.

It may happen that a single muon is reconstructed incorrectly as two muons or even more. To cope with this situation, additional “ghost-cleaning” is performed. Those tracker Muons which are not Global Muons, are needed to be arbitrated. If two muons are sharing their segments upto 50% or more, lower quality muon is removed.

1.3.2 Muon Isolation Requirement

Muon isolation used is based on Particle-Flow. So-called $\Delta\beta$ correction is used in order to subtract the pileup contribution for muons, whereby $\Delta\beta = \frac{1}{2} \sum_{\text{PU}}^{\text{charged had.}} p_T$

TABLE 1.3: The requirements for a muon to pass the Tracker High- p_T ID. Note that these are equivalent to the Muon POG High- p_T ID with the global track requirements removed.

Plain-text description	Technical description
Muon station matching	Matching of muon to the segments in two muon stations at least NB: It means that muon is an arbitrated tracker muon.
Good p_T measurement	$\frac{p_T}{\sigma_{p_T}} < 0.3$
Vertex compatibility ($x - y$)	$d_{xy} < 2$ mm
Vertex compatibility (z)	$d_z < 5$ mm
Pixel hits	One pixel hit at least
Tracker hits	Hits in at least six tracker layers

gives an estimate of the energy deposit of neutral particles (i.e. photons and hadrons) from the pile-up vertices. Relative isolation for muons is then defined as:

$$\text{RelPFiso} = \frac{\sum^{\text{charged had.}} p_T + \max(\sum^{\text{neutral had.}} + \sum^{\text{photon}} - \Delta\beta, 0)}{p_T^{\text{lepton}}} \quad (1.1)$$

The isolation working point for muons was optimized in Ref. [2] and the working point was chosen to be the same as electrons, namely $\text{RelPFiso}(\Delta R = 0.3) < 0.35$.

1.3.3 Muon Impact Parameter Selection

Selection on muon significance of impact parameter(SIP) is used same as for the electrons which is described in Section 1.2.3:

- $|\text{SIP}_{3D} = \frac{\text{IP}}{\sigma_{\text{IP}}}| < 4$

1.3.4 Muon Energy Calibrations

Similar to electrons the momentum scale of muon is measured in data by fitting a Crystall-ball function to the di-muon invariant mass spectrum around the Z peak in the $Z \rightarrow \mu\mu$ control region. From Fig. 1.12 shows a very good comparison of data with simulation.

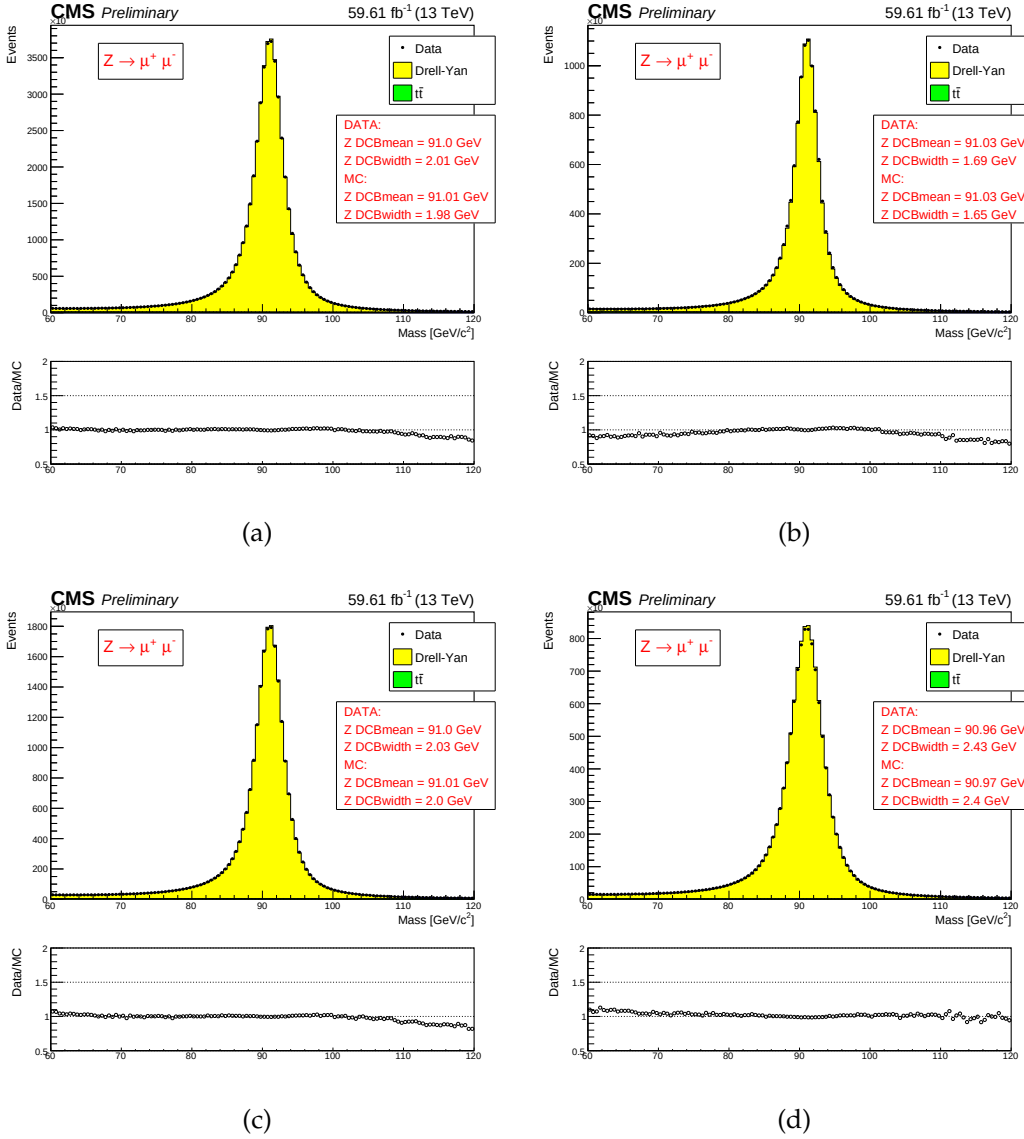


FIGURE 1.12: (a): muon energy scale measured in the $Z \rightarrow \mu\mu$ control region for all muons, for both of muons from barrel region (b), for one muon from barrel, one from endcaps (c) and for both muons in the endcaps (d). The results in the figures are extrated from fits using the Crystall-ball function.

1.3.5 Efficiency Measurements of muons

Efficiencies of muons are measured using same method used for electron measurements i.e. Tag-and-Probe (T&P) method performed on $Z \rightarrow \mu\mu$ and $J/\psi \rightarrow \mu\mu$ events in bins of p_T and η . More details on the methodology are described in Section 1.1.1 and also in Ref. [3]. Measurements are extracted using 2018 RunA,B,C,D

data while the measurements corresponding to 2016 and 2017 datasets have already been reported in Ref. [4] and Ref. [5] respectively. For the measurement of muon reconstruction and identification efficiency at high p_T ($p_T > 20 \text{ GeV}$), and impact parameter requirements and isolation efficiencies for all p_T , Z sample is used. To measure the muon reconstruction and identification efficiency at high p_T , and the efficiency of the isolation and impact parameter requirements at all p_T . To measure the reconstruction and identification efficiency at low p_T , J/ψ sample is used as it benefits from a better purity in that kinematic regime. In this case, events are collected using `HLT_Mu7p5_Track2_Jpsi_v*` when probing the reconstruction and identification efficiency in the muon system, and using the `HLT_Mu7p5_L2Mu2_Jpsi_v*` when probing the tracking efficiency.

Muon reconstruction and identification efficiency measurement Results for muon reconstruction and identification efficiencies for high p_T muons ($p_T > 20 \text{ GeV}$) have been derived by the Muon POG. The probe in this measurement are tracks of inner tracker, and the passing probes are those which are also regarded as a global or tracker muon and passing the Muon POG Loose muon identification. Results for low p_T muons were derived using events, with the same definitions of probe and passing probes. The systematic uncertainties on the measurements are estimated by including several variations e.g. varying the analytical signal and background shape models used to fit distribution of dimuon invariant mass. Furthermore, number of bins in dimuon distribution and mass window are also increased and decreased to include systematic effects. Details on the procedure can be found in Ref. [2]. The efficiency and scale factors used for low p_T muons are the ones derived using single muon prompt-reco dataset.

The data and simulation efficiencies and their ratios in bins are shown in Fig. 1.13.

Efficiency measurement of muon impact parameter requirement The measurement is performed using Z events. Events are selected with `HLT_IsoMu27_v*` or `HLT_Mu50_v*` triggers. For this measurement, the probe is a muon passing the POG Loose identification criteria, and it is considered a passing probe if satisfies the SIP3D, dxy, dz cuts of this analysis. The results are shown in the Fig. 1.14.

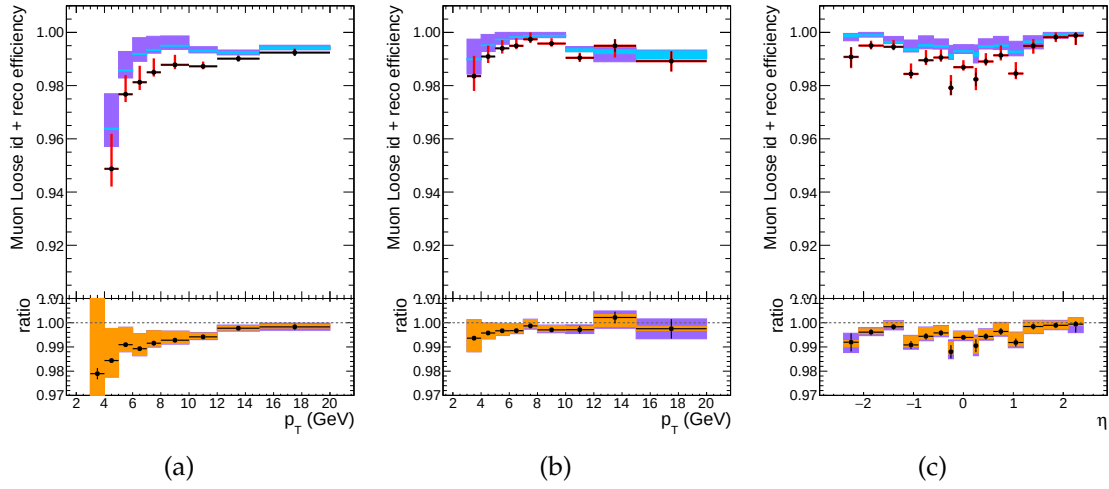


FIGURE 1.13: Reconstruction and identification efficiencies of muons at low p_T , measured with the $T\&P$ method on events, in bins of p_T in barrel (left) and endcaps (center), and in terms of η for $p_T > 7$ GeV (right). In upper panel of each diagram, the larger uncertainty bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of two efficiencies, black uncertainty bars are for statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

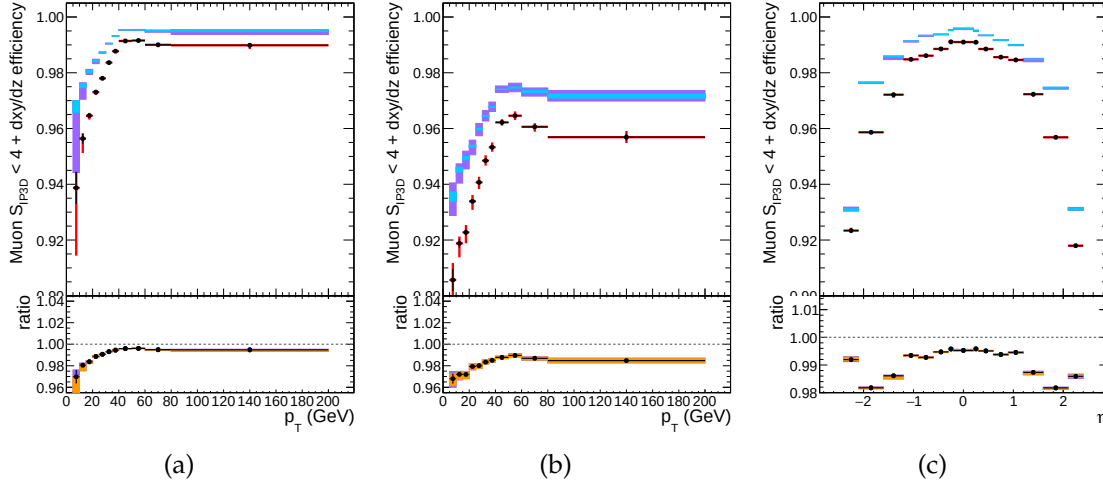


FIGURE 1.14: Efficiency of the muon impact parameter requirements, measured with the $T\&P$ method on Z events, in bins of p_T in barrel (left) and endcaps (center), and in terms of η for all p_T (right). In upper panel, the larger uncertainty bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of two efficiencies, black error bars are for statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

Efficiency measurement of muon isolation requirements Isolation efficiency is measured using events from the Z decay for any p_T . The events are selected with either of `HLT_IsoMu27_v*` or `HLT_Mu50_v*` triggers (for 2017 `HLT_IsoMu27_v*` is used and for 2016 measurements, *OR* condition is required among `HLT_IsoMu20_v*`, `HLT_IsoTkMu20_v*`, `HLT_IsoMu22_v*`, `HLT_IsoTkMu22_v*`, `HLT_IsoMu24_v*`, `HLT_IsoTkMu24_v*`). The isolation of the muons are calculated after recovery of the FSR photons and subtracting their contribution to the isolation cone of the muons. There are details described of this method which can be found in Ref. [6].

The results are shown in Fig. 1.15.

Tracking Tracking efficiency (i.e. efficiency to associate a track to some muon in inner detector) is almost 100% as shown in the Figure. 1.1 and hence muon Physics Object Group no more recommends tracking scale factors in the analyses.

Overall results of muon Scale Factors The product of all scale factors for reconstruction, identification, impact parameter and isolation requirements is shown in

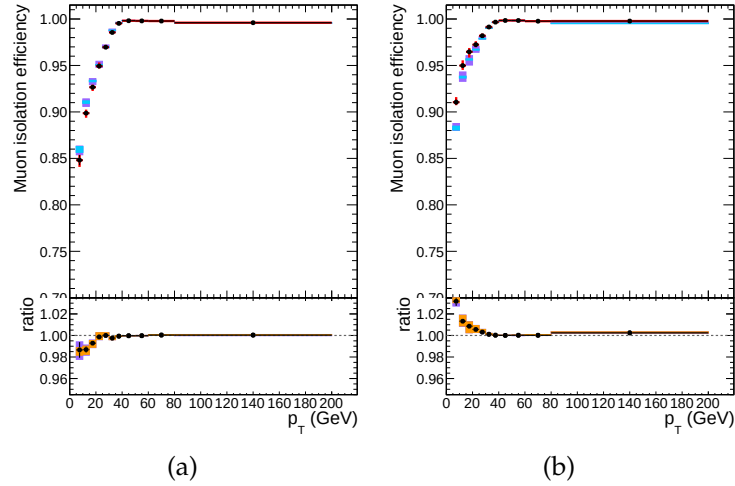


FIGURE 1.15: Efficiency of muon isolation requirement, measured with the tag&probe method on Zevents, as function of p_T in the barrel (left) and endcaps (right). In the upper panel, the larger error bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of the two efficiencies, the black error bars are for the statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

434 two dimensional display in Fig. 1.16. The figure also shows overall uncertainty on
 435 scale factors.

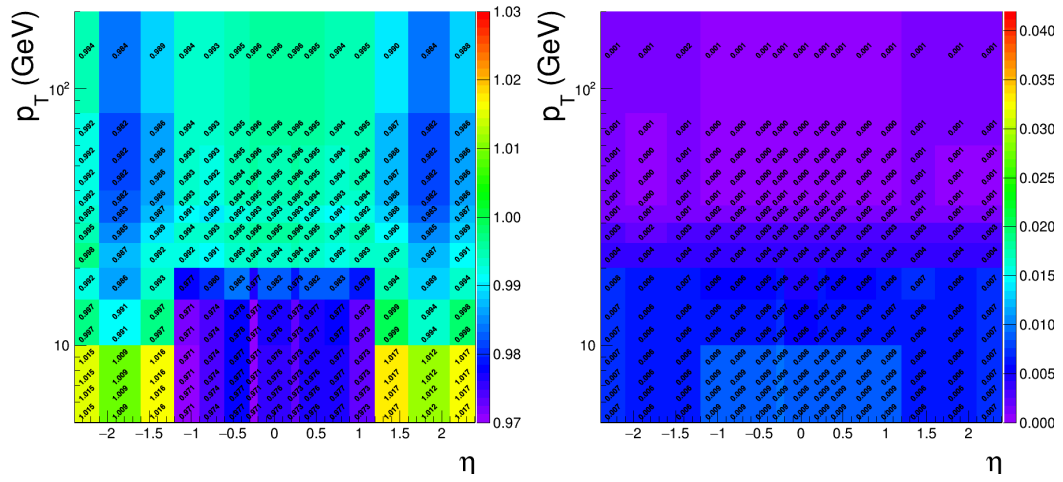


FIGURE 1.16: Left: Overall scale factors for muons, as function of p_T and η . Right: Uncertainties on overall scale factors muons shown in two dimensions of p_T and η .

1.4 Photons for FSR recovery

The procedure of FSR recovery is described as under:

1. Photons are initially collected using PF algorithm which are called PF photons.
2. These photons are preselected by requiring $p_{T,\gamma} > 2 \text{ GeV}$, $|\eta^\gamma| < 2.4$, and a relative Particle-flow isolation smaller than 1.8 (computed within a cone of radius $R = 0.3$).
3. Supercluster (SC) veto: All photons matching (direct association with PF candidates) with some electron that passes ID and SIP requirements are removed.
4. Association of photons is made with lepton which is closest and passes ID and SIP requirements.
5. Photons should fulfill the requirements of $\Delta R(\gamma, l)/E_{T,\gamma}^2 < 0.012$, and $\Delta R(\gamma, l) < 0.5$.
6. For a lepton, a photon with least $\Delta R(\gamma, l)/E_{T,\gamma}^2$ is considered.
7. FSR photons are excluded from sum of isolation of all leptons in an event which pass both loose ID and SIP requirements.

More details on the optimization of the FSR photon selection can be found in Ref. [2, 6].

1.5 Jets

In a high energy experiment e.g. proton-proton collision at LHC, jets are produced by quarks and/or gluons when they go through the process of hadronization and making a cone like structure in the detector like CMS. Hence, jets are physics objects which provide the experimental evidence of quarks and the gluons. As these jets of particles propagate through the CMS detector, they leave signals in components such as the tracker and the electromagnetic and hadronic calorimeters. A typical diagram of the a jet can be seen in the Figure. 1.17 and in Ref. [jets]

Information from these signatures is exploited through some jet algorithms to fully reconstruct a “jet”.

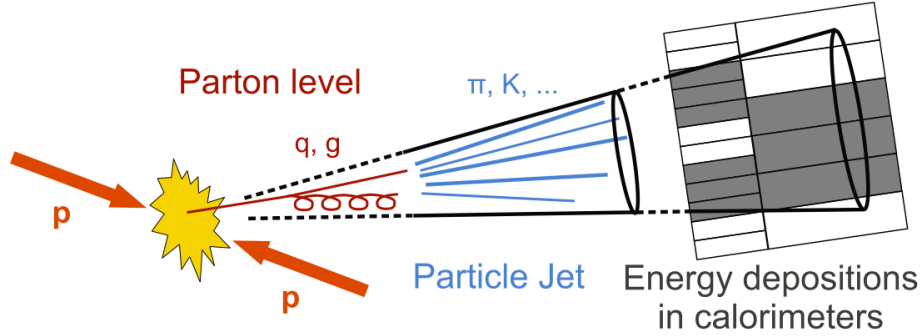


FIGURE 1.17: A schematic diagram of proton-proton collision and signature for a jet cone.

The processes of Higgs production through fusion of vector bosons (VBF) and through other mechanisms normally differ as regards the jet kinematics.

1.5.1 Jet Identification

For the analysis upon which the dissertation is built, jets are reconstructed through anti- k_T clustering algorithm out of particle-flow candidates, with distance parameter $R = 0.4$, after rejecting the charged hadrons that are associated with a pileup primary vertex.

To reduce instrumental background, the tight working point jet ID suggested by the JetMET Physics Object Group is applied [7]. In addition, jets from Pile-Up are rejected using the PileUp jet ID criteria suggested by the JetMET POG [8]¹. In this analysis, the jets from within $|\eta| < 4.7$ area are required to have a transverse momentum greater than 30 GeV. In addition, jets are cleaned from any of the tight leptons (passing the SIP and isolation cut computed after FSR correction) and FSR photons by a separation criterion: $\Delta R(\text{jet}, \text{lepton}/\text{photon}) > 0.4$.

¹In 2016, jet pileup ID is not applied (to be included in upcoming manuscript on the same subject).

1.5.2 Jet Energy Corrections

Energy of the jets as measured from calorimeters is not exactly what is true energy at hadron level. There are many experimental reasons, e.g non-linear response of calorimeters, addition of energies from the pileup events, possible mixing from non-instrumented regions or noise in the detector, due to which the true energy may not be associated to the jets and hence, one needs to estimate the corrections to energy of a jet. These corrections are to be applied both in data and the simulations to match with the true hadron level energy.

In the analysis, standard jet energy corrections are implemented to the reconstructed jets, which consist of L1 Pileup, L2 Relative Jet Correction, L3 Absolute Jet Correction for both Monte Carlo samples and data, and also residual calibration for data [9]. Jet Energy resolution corrections are also applied to MC.

1.5.3 B-tagging

B-hadrons or b-jets have long life time *viz.* about $1.5ps$ (those of c-hadrons have $1ps$ or even less), and travel longer in the detector before they decay and subsequently leave the tracks of large impact parameter depending upon the momentum. These tracks are called displaced tracks and can help to reconstruct the secondary vertex (SV). A typical diagram of a heavy-flavour jet with a SV from the decay of b or c hadron resulting in charged-particle tracks is shown in the Figure. 1.18 and in Ref. [bjet].

In order to distinguish whether a jet is b-jet or not *DeepCSV* algorithm is used as our b-tagging algorithm. It combines the same information as the previous tagger *CSVv2* i.e. SIP, jet kinematics and secondary vertex but uses information of more tracks. Also, the b-tag output discriminator is computed with a Deep Neural Network. In the analysis, a jet is considered as a b-tagged if it meets requirement of *medium* working point, i.e. if its `|pfDeepCSVJetTags:probb+ pfDeepCSVJetTags:probbb|` discriminator is greater than 0.4184 [10].

Scale factors of data to simulation efficiencies for b-tagging are provided for this working point for the full dataset varied in jet p_T , η and flavour. They are applied to simulated jets by downgrading (upgrading) the b-tagging status of a fraction of the b-tagged (untagged) jets that have a scale factor smaller (larger) than one.

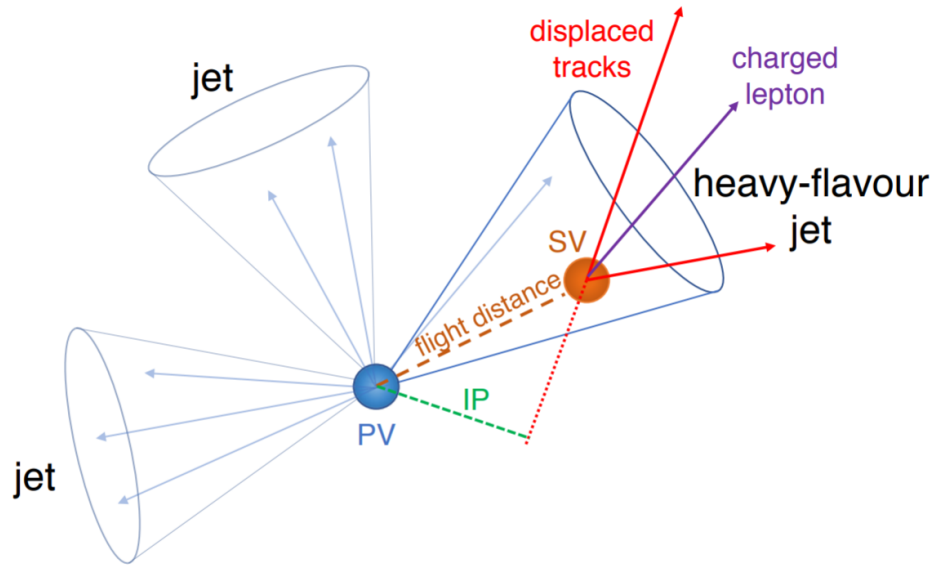


FIGURE 1.18: A schematic diagram of a heavy flavor jet with a secondary vertex.

1.5.4 Validation on data

Some validation studies are performed to probe agreement of data with simulated samples for jets. It is similar to what was done with electrons and muons. Events containing or $Z \rightarrow \mu^+ \mu^-$ were selecting, following the same trigger and lepton selection criteria of the main analysis, but increasing the p_T threshold on leptons to 30 GeV and asking the dilepton invariant mass window of 70-110 GeV. Basic properties of additional jets in an event, requiring $p_T > 30$ GeV and $\eta < 4.7$, were then compared. Figure 1.19 shows comparison of data with the MC in bins of pseudo-rapidity of the leading jet. Simulated samples of Drell-Yan and $\bar{\nu}\nu$ are used. Uncertainties from the jet energy corrections are also displayed on ratio plots of data to MC.

The disagreement of data with MC for the MET distribution prevents us at this stage to use this variable in the analysis, in particular to better target VH events where, for instance, a Z would decay into neutrinos.

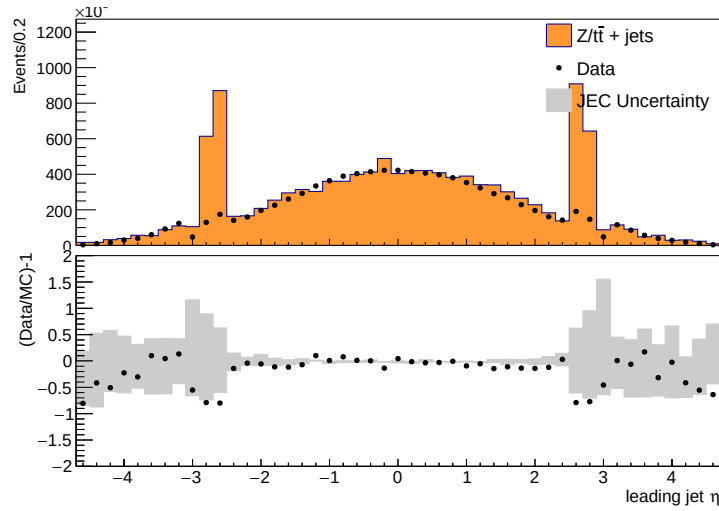


FIGURE 1.19: Comparison of data with MC for leading jet η in $Z \rightarrow \ell\ell$ +jets. MC is normalized to data. Jet ID and Jet PUID are applied. MC samples include DY and $t\bar{t}$. Data/MC ratio plots are shown in the bottom of each plots, together with the uncertainties (shaded histograms) from Jet Energy Corrections.

1.6 Object Summary

The requirements on all objects used in the analysis (with 2018 data) are summed up in the Table 1.4. In addition, a “ghost-cleaning” procedure is applied to the muons, as described in Sec. 1.3.1.

A lepton is declared **loose** if it passes the reconstruction, kinematics and dxy/dz cuts and declared **tight** if it passes in addition the identification, isolation and SIP3D cut.

For 2016 and 2017 overall selections are same except a few as given below:

- For 2018 and also for 2017, BDT includes ID and as well as Isolation requirements; however, for 2016 BDT includes only ID and cut based Isolation is explicitly applied on the top of BDT.
- In 2017 and 2018, *DeepCSV* is used as B-tagging algorithm. For 2016, *CSVv2* was used as B-tagging algorithm (with chosen working point as `pfCombinedInclusiveSecondaryVertexV2BJetTags` discriminator should be greater than 0.8484)

TABLE 1.4: Chart of selection of physics objects for the $H \rightarrow ZZ \rightarrow 4\ell$ analysis used on the 2018 data (few differences w.r.t 2016 and 2017, as described in Section. 1.6 and rest of selections remain same in all years.)

Electrons
$p_T^e > 7 \text{ GeV}$ and $\eta^e < 2.5$ $d_{xy} < 0.5 \text{ cm}$ and $d_z < 1 \text{ cm}$ $ SIP_{3D} < 4$ BDT ID with isolation (Fall17v2 training) with cuts from Table 1.2
Muons
Global or Tracker Muon Discard Standalone Muon tracks if reconstructed in muon system only Discard muons with muonBestTrackType==2 even if they are global or tracker muons $p_T^\mu > 5 \text{ GeV}$ and $\eta^\mu < 2.4$ $d_{xy} < 0.5 \text{ cm}$ and $d_z < 1 \text{ cm}$ $ SIP_{3D} < 4$ The PF muon ID if $p_T < 200 \text{ GeV}$, PF muon ID or High- p_T muon ID (Table 1.3) if $p_T > 200 \text{ GeV}$ $\mathcal{I}_{PF}^\mu < 0.35$
The FSR photons
$p_T^\gamma > 2 \text{ GeV}$ and $\eta^\gamma < 2.4$ $\mathcal{I}_{PF}^\gamma < 1.8$ $\Delta R(\ell, \gamma) < 0.5$ and $\frac{\Delta R(\ell, \gamma)}{(p_T^\gamma)^2} < 0.012 \text{ GeV}^{-2}$
Jets
$p_T^{\text{jet}} > 30 \text{ GeV}$ and $\eta^{\text{jet}} < 4.7$ $\Delta R(\ell/\gamma, \text{jet}) > 0.4$ Cut-based jet ID (tight WP) Jet pileup ID (tight WP) Deep CSV algorithm for b-tagging (medium WP)

Bibliography

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- [1] Vardan Khachatryan et al. "Performance of electron reconstruction and selection with the CMS detector in proton-proton collisions at $\sqrt{s} = 8$ TeV". In: *JINST* 10 (2015), P06005. DOI: [10.1088/1748-0221/10/06/P06005](https://doi.org/10.1088/1748-0221/10/06/P06005).
- [2] *Measurement of the properties of the Higgs boson in the four-lepton final state at $\sqrt{s} = 13$ TeV*. CMS Physics Analysis Note CMS-AN-15-277. 2016.
- [3] *Measurement of the properties of the Higgs boson in the four-lepton final state at $\sqrt{s} = 13$ TeV*. CMS Physics Analysis Note CMS-AN-15-277. 2016.
- [4] R. Salerno and M. Ahmad. "Measurements of properties of the Higgs boson in the four-lepton final state at $\sqrt{s}=13$ TeV". In: *CMS AN* 2016/442 (2016).
- [5] T. Sculac and D. Sperka. "Measurements of properties of the Higgs boson in the four-lepton final state at $\sqrt{s}=13$ TeV with 2017 data". In: *CMS AN* 2017/342 (2017).
- [6] *New results on the study of Higgs boson production in the four-lepton final state at $\sqrt{s} = 13$ TeV*. CMS Physics Analysis Note CMS-AN-16-217. 2016.
- [7] JetMETPOG. "Jet ID for 2018". In: *CMS Twiki* (2018). URL: <https://twiki.cern.ch/twiki/bin/view/CMS/JetID13TeVRun2018>.
- [8] JetMETPOG. "Jet PU ID for 2017". In: *CMS Twiki* (2017). URL: https://twiki.cern.ch/twiki/bin/viewauth/CMS/PileupJetID#Information_for_13_TeV_data_anal.
- [9] JetMETPOG. "JEC MC for 2018". In: *CMS Twiki* (2018). URL: https://twiki.cern.ch/twiki/bin/view/CMS/JECDataMC#2018_Data.
- [10] BTAGPOG. "B-tagging for 2018". In: *CMS Twiki* (2018). URL: <https://twiki.cern.ch/twiki/bin/viewauth/CMS/BtagRecommendation102X>.